

Lirui Wang

5000 25th Ave NE, Seattle, WA 98105

✉ liruiw@uw.edu

🌐 liruiw.github.io

☎ +1 (206) 483-8515

EDUCATION

University of Washington, Seattle, WA

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

Master of Science, Computer Science, College of Arts and Sciences

Sep. 2016 – Present.

GPA: 3.97 / 4.00, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Seattle, WA

UW Robotics and States Estimation Lab

March. 2019 – Present.

- Implemented CHOMP and introduced an integrate planner that optimizes manipulation trajectory with online grasp synthesis and goal selection.

Survey and Empirical Evaluation of Attacks and Defense for Robust Classification, Seattle, WA

Robust Deep Learning Course Project

Sep. 2019 – Dec. 2019

- Investigate example reweighting and adversarial training techniques to robustify a fossil classification task and survey recent papers about attacks and defense of adversarial examples for neural networks.

Machine Learning in Robot Trajectory Optimization, Seattle, WA

Statistical Learning Course Project

Sep. 2019 – Dec. 2019

- Summarized the formulation of trajectory optimization as functional gradient descent using RKHS and as probabilistic inference using GP.

6D Object Pose Tracking with the Deep Iterative Matching Network, Seattle, WA

UW Robotics and States Estimation Lab

June. 2018 – March. 2019

- Extended DeepIM to robustly track robot arm and YCB objects for occluded scenes and AR applications.

Unsupervised Flow and Depth Methods Study, Seattle, WA

UW Robotics and States Estimation Lab

Jan. 2018 – Jun. 2018

- Investigated unsupervised learning methods such as SfMlearner and MonoDepth for 3D geometry tasks such as depths, flow, and egomotion on a robotic setting
-

EXPERIENCE

Research Intern, NVIDIA, WA

Jun. 2020 – Sep. 2020

Industry Capstone Project, Microsoft, WA

Jan. 2018 – Present.

- Built on autonomous mobile manipulator to pull out servers from server rack and transport it to repair.

Teaching Assistant, University of Washington, WA

Sep. 2019 – Mar. 2020

- Help instructors teach CSE 455 Computer Vision and CSE 473 Artificial Intelligence.

Research Assistant, UW Robotics and State Estimation Lab, WA

Sep. 2018 – Present.

- Worked on manipulation research projects with Dr. Yu Xiang and Prof. Dieter Fox

Robotics Engineer Intern, DJI Technology, China

Jun. 2017 – Aug. 2017

- Worked on an automatic robot battle that composes of a vehicle navigated by SLAM and a DJI Manifold with Guidance
-

PUBLICATIONS

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Lirui Wang, Yu Xiang, Dieter Fox

In Submission of *Robotics: Science and Systems* (RSS 2020)

SKILLS & OTHERS

Programming Language: Java, C++/C, Python, Matlab

Tools: ROS, Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, SLAM

Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing, Statistical Learning, High-Dimension Statistics, Robust Machine Learning, Reinforcement Learning, Interactive Learning

Scholarships: Tau Beta Pi Scholarship 2017-2019, Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship 2017-2018, Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship 2018-2019